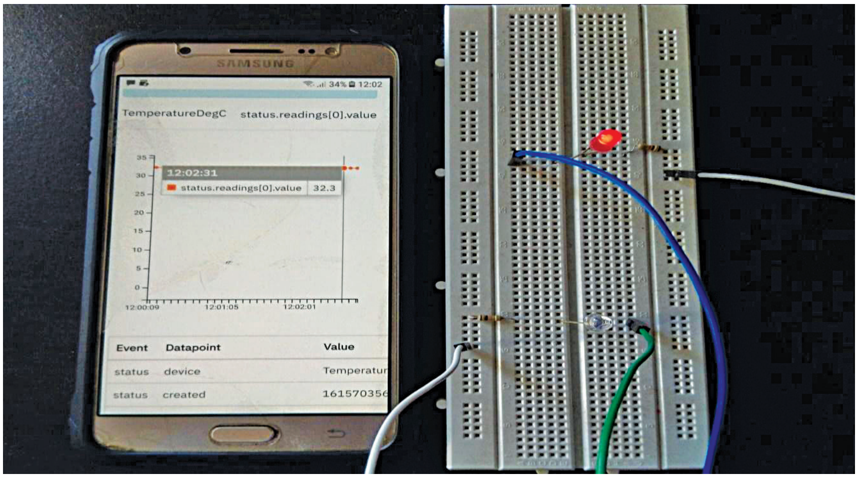

Fig. 10: Red LED actuation with corresponding values

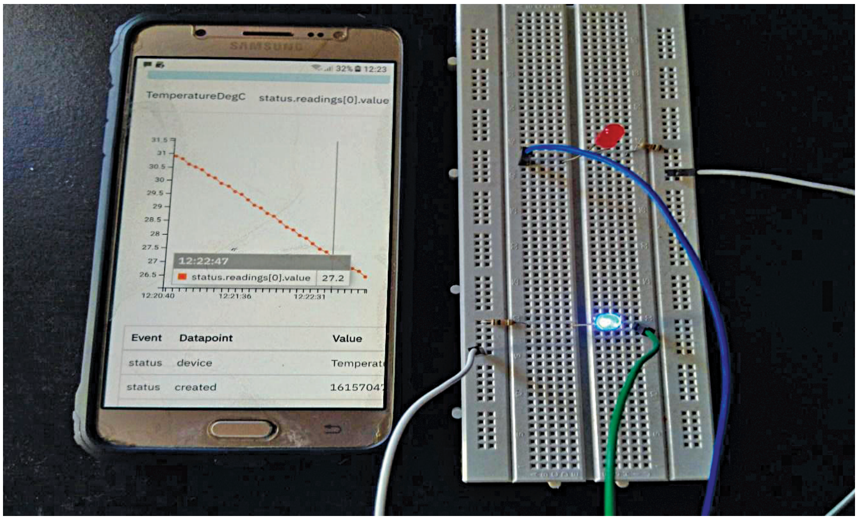

Fig. 11: Blue LED actuation with corresponding values

msg="Sent data to MQTT Broker"

level=DEBUG ts=2021-03-21T14:35:48.37480595Z app=AppService- source=context.go:80 msg="Marking event as pushed"

Snapshots of sensor data from IBM Cloud portal (*https://internetofthings.ibmcloud.com*) are shown in Fig. 9.

Rules engine

Another application service sends the data to rules engine Kuiper, which actuates the LEDs by triggering the corresponding GPIO core commands when configured temperature thresholds are crossed. Kuiper receives the sensor data sent by the application service and applies the rules which are preconfigured for actuation.

For the case study, following are the steps required to configure rules in Kuiper:

----

bin/kuiper drop stream edgex_data

bin/kuiper create stream edgex_data"() WITH(FORMAT=\"JSON\", TYPE=\"edgex\")"

bin/kuiper create rule red_led_on -f rule1.txt

bin/kuiper create rule red_led_off -f rule2.txt

bin/kuiper create rule blue_led_off -f rule3.txt

bin/kuiper create rule blue_led_on -f rule4.txt

From the above, rule1.txt actuates red LED on when temperature value goes beyond 30 degrees, and rule2.txt actuates red LED off when temperature drops below 30 degrees. Similarly, rule3.txt actuates blue LED off when temperature goes beyond 28 degrees, and rule4.txt actuates blue LED on when temperature drops below 28 degrees. Logs when red LED is actuated are:

----

root@ubuntu:/home/ubuntu/new_testing/kuiper/kuiper/_build/kuiper-1.1.1-4-g24892eb-linux-aarch64# tail -f ./log/stream.log

time="2021-03-02 14:35:21" level=info msg="sink result for rule blue_led_off: [{\"TemperatureDegC\":33.6}]" file="sinks/log_sink.go:16" rule=blue_led_off

time="2021-03-02 14:35:21" level=info msg="sink result for rule red_led_on: [{\"TemperatureDegC\":33.6}]" file="sinks/log_sink.go:16" rule=red_led_on

----

Instead of LED actuation, necessary actions can be taken for industrial use cases through relays and solenoids.

GPIO device service

GPIO device service switches the LEDs on/off when core command of corresponding GPIO pin is triggered by Kuiper. A snapshot of the red LED actuation along with live values in IBM Cloud portal are shown in Fig. 10. Fig. 11 shows a snapshot of the blue LED's actuation with corresponding values. EFY

Vijay Kumar Gopu, working as a technical specialist with HCL Technologies, has working experience in the areas of Virtualisation and Wireless Core Networks